

QUICK SORT:

Quick Sort is a comparison-based sorting algorithm that selects an element called a pivot and partitions the array so that smaller elements are placed on one side and larger elements on the other side.

ALGORITHM:

1. Select an element as the pivot.
2. Compare the other elements with the pivot.
3. Place smaller elements on the left side.
4. Place larger elements on the right side.
5. Put the pivot in its correct position.
6. Apply Quick Sort recursively to the left part.
7. Apply Quick Sort recursively to the right part.
8. Continue until the array is sorted.

Time Complexity:

Case	Complexity
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Best	$O(n \log n)$
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Average	$O(n \log n)$
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Worst	$O(n^2)$
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Space Complexity:

Space Complexity = $O(n)$

OUTPUT:

```
27 void quickSort(int arr[],int low,int high){
Enter the number of elements: 4
Enter the element 1: 44
Enter the element 2: 25
Enter the element 3: 4
Enter the element 4: 65
Sorted array: 4 25 44 65

...Program finished with exit code 0
Press ENTER to exit console.
```

CONCLUSION:

Quick Sort is generally very fast and efficient in practice. Its average time complexity is **$O(n \log n)$** , but choosing a poor pivot can result in **$O(n^2)$** in the worst case.